Deployment Hardware Guide for Data Collection

Bill of Materials

Part	Quantity	Link	Price (per unit)
Modified UMI handheld device	2	-	€624.72
HTC VIVE Base Station	2	Link	€245.00
Tripod	2	Link	€39.99
GoPro Hero 9 (head-mounted & accessories)	1	Link Link	€179.99 €49.99
Mobile phone for GoPro monitoring/pairing	1	-	
Laptop / workstation	1	-	
Microphone Rode Gen 3	1	Link	€235.90
Chargers and charging cables	as needed	-	
Extension cables	as needed	-	
Power adapters	as needed	-	
Spare parts for modified UMI	as needed	-	
Spare soft fingers / hard fingers	as needed	-	

Setup Instruction

1. Prepare the workspace.
2. Position the two VIVE base stations.
3. Assign different base-station channels.
4. Launch SteamVR with the required null-driver configuration.
5. Synchronize the workstation time.
6. Power on and pair the VIVE trackers.
7. Mount the trackers on the modified UMI devices.

8. Perform room and tracker calibration using the 17 cm tracker offset reference.
9. Power on and pair the mounted GoPro Hero 13 cameras.
10. Prepare the optional head-mounted GoPro Hero 9.
11. Prepare the microphone.
12. Launch vilma_data_collection.py.
13. Select unimanual/bimanual mode and optional head-mounted camera mode.
14. Confirm all device connections.
15. Start recording.
16. Execute the task.
17. Stop recording and save the run.
18. Record metadata and inspect tracker outliers.
19. Inspect the handheld hardware and replace parts if needed.
20. Reset the setup for the next run.